What we fingerprint, and the two questions we ask

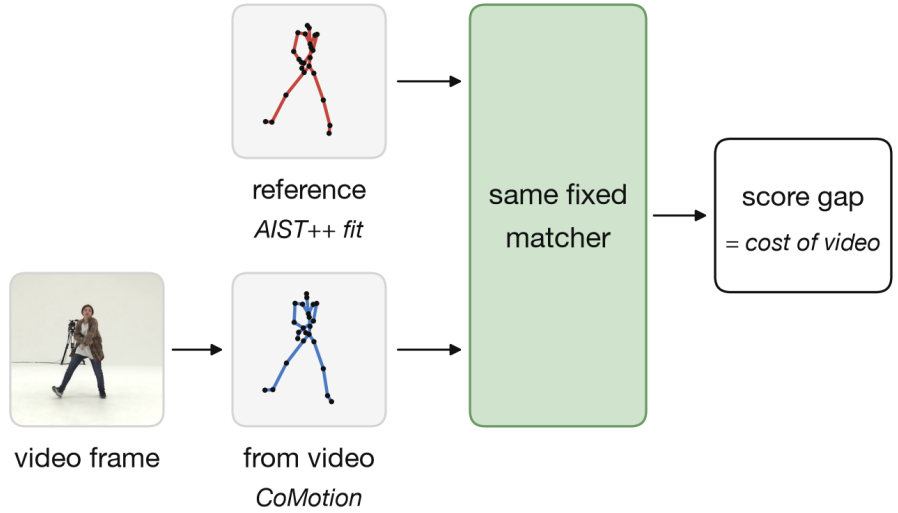
THE TASK · RANK THE DATABASE

Decision: **gLH_sBM_ch01** · best score 0.4911 vs threshold ξ = 0.4575 · true label gLH_sBM_ch01, 11 relevant references

#	reference	label	score	ℓ*	tempo	
1	gLH_sBM_cAll_d16_mLH1_ch01	gLH_sBM_ch01	0.4911	0	1.12×	✓
2	gLH_sBM_cAll_d16_mLH2_ch01	gLH_sBM_ch01	0.4644	0	1.25×	✓
3	gLH_sBM_cAll_d16_mLH0_ch02	gLH_sBM_ch02	0.4453	-18	1.00×	
4	gLH_sBM_cAll_d16_mLH1_ch02	gLH_sBM_ch02	0.3880	-8	1.12×	
5	gLH_sBM_cAll_d18_mLH2_ch02	gLH_sBM_ch02	0.3305	7	1.25×	
6	gLH_sBM_cAll_d18_mLH4_ch02	gLH_sBM_ch02	0.3189	12	1.40×	
7	gLH_sBM_cAll_d16_mLH3_ch01	gLH_sBM_ch01	0.3162	0	1.40×	✓

Our web interface: the same choreography (✓) should rank first

QUESTION 1 · IS THE SKELETON FROM VIDEO GOOD ENOUGH?



QUESTION 2 · CAN WE TELL THAT A VIDEO IS AI-GENERATED?

LEVEL 1

No library of real clips: judge the motion alone

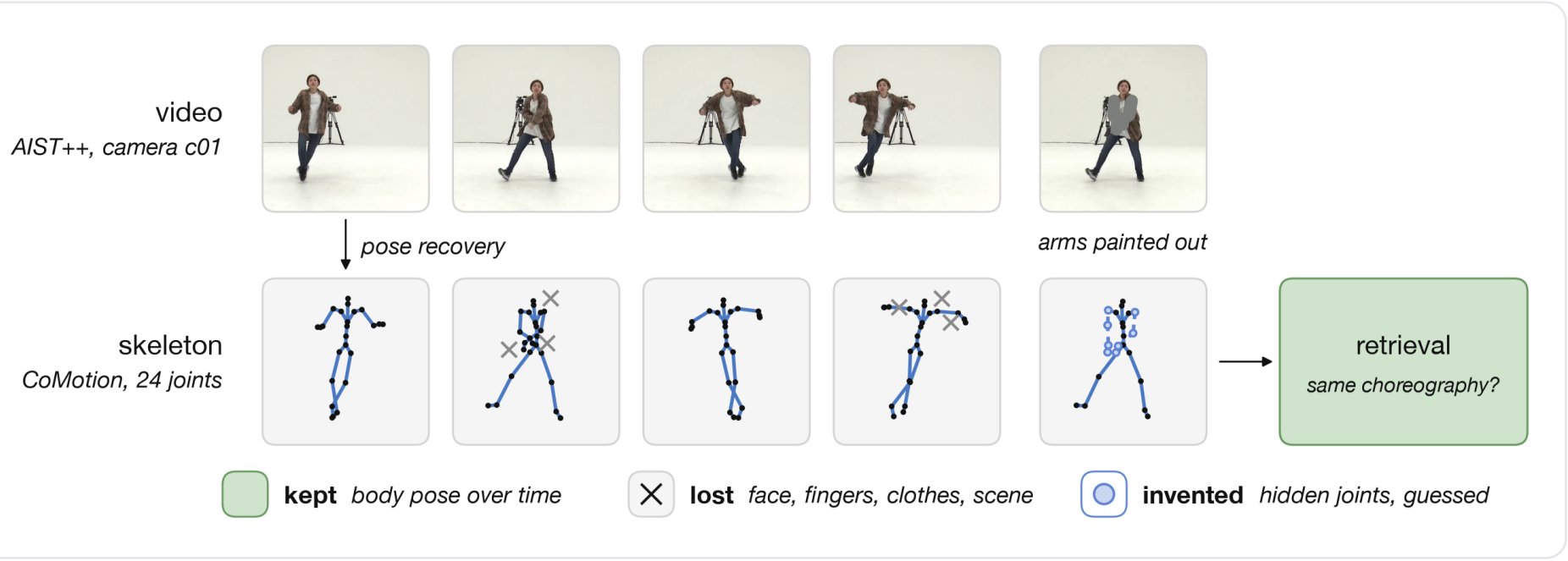
LEVEL 2

With a library of real clips: does the video copy one?

Arina Kharlamova · Advanced Computer Vision, MBZUAI · Mentor: Mohammed Irfan K · 14 October 2026

FROM PROFESSOR · the pose-recovery literature was missing

A skeleton keeps pose over time, loses hands and face, and invents hidden joints



POSE RECOVERY
HMR · 4DHumans · WHAM · GVHMR · CoMotion

RETRIEVAL
Choreo · TMR · MotionBERT · Pr-VIPE · DTW · ST-GCN

GENERATION
Animate Anyone · MimicMotion · Wan-Animate

SETUP · what we use

Data and models used

DATA

- **AIST++** choreography, 9 cameras, reference skeletons
- **NTU RGB+D** everyday actions
- **Cross-camera videos** ~164; animation corpus 230
- **Social media dataset** real performances and copies
- **Generated** Wan-Animate copies (36), text-to-video (24), MimicMotion

MODELS

- **CoMotion** frozen: video → skeleton + 2D confidence
- **AIST++ fits** geometry supervision
- **kNN, DTW** fixed matchers
- **ChoreoMatch** our reproduction of Choreo
- **Ours** refiner + part encoder + visibility, InfoNCE
- **Generators** Wan-Animate, MimicMotion



Our web interface: query (blue), best match (orange), agreement per time slice

STATUS · since the proposal

Status: five experiments on AIST++ are complete; the side-view retraining and three comparisons are open

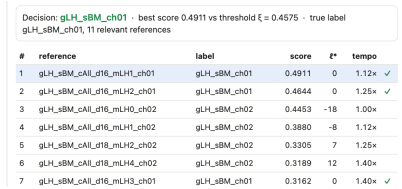
1 Skeletons from video



4–6 cm

pose error on 9 cameras; 11 cm with the arms hidden

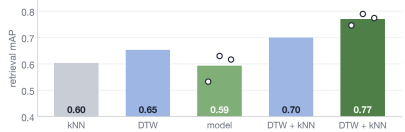
2 Matching



3 matchers

kNN, DTW and our ChoreoMatch behind one web interface

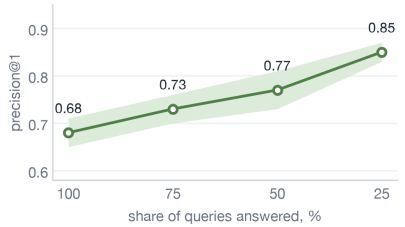
3 Trained model



0.70 → 0.77

mAP when the model joins DTW + kNN, in all 3 seeds

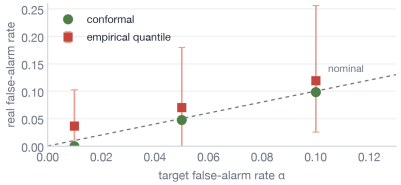
4 Where the detail lives



0.99 → 0.30

true match after one arm is swapped; 0.68 → 0.85 if we answer only when sure

5 AI-copy provenance



AUC 0.96–0.997

with a library of real clips; conformal threshold holds the nominal false-alarm rate

STILL RUNNING AND STILL OPEN

IN PROGRESS

Side view

retrain with the full yaw range, 2 of 3 seeds done

OPEN

Offline estimator

WHAM, GVHMR against CoMotion

OPEN

Direct-video baseline

is a skeleton needed at all?

OPEN

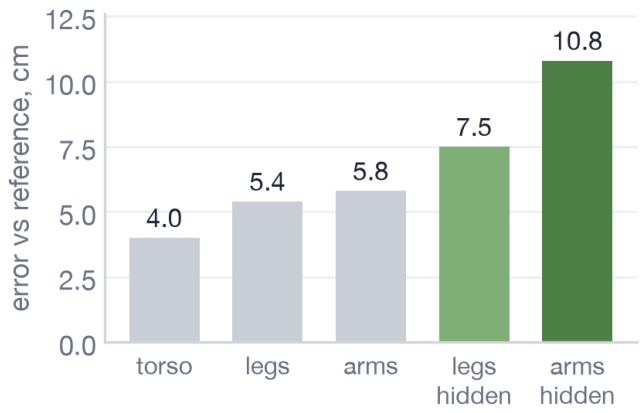
Social media dataset

threshold needs 99+ real clips for 1 %

RESULT 1 · what the video step costs

Pose-estimation error is small on front views and grows on the side view and for hidden limbs

POSE ERROR BY BODY PART



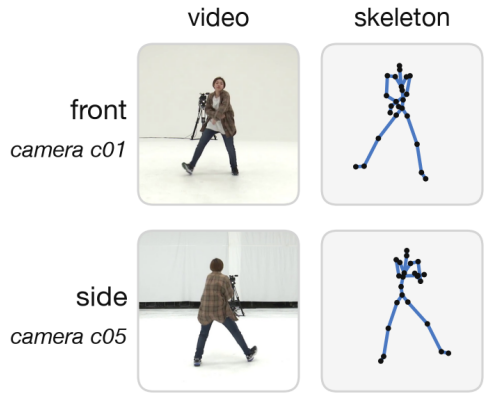
hidden = painted out of the video

RETRIEVAL MAP BY CAMERA



same matcher, reference vs CoMotion skeletons

THE SAME MOMENT

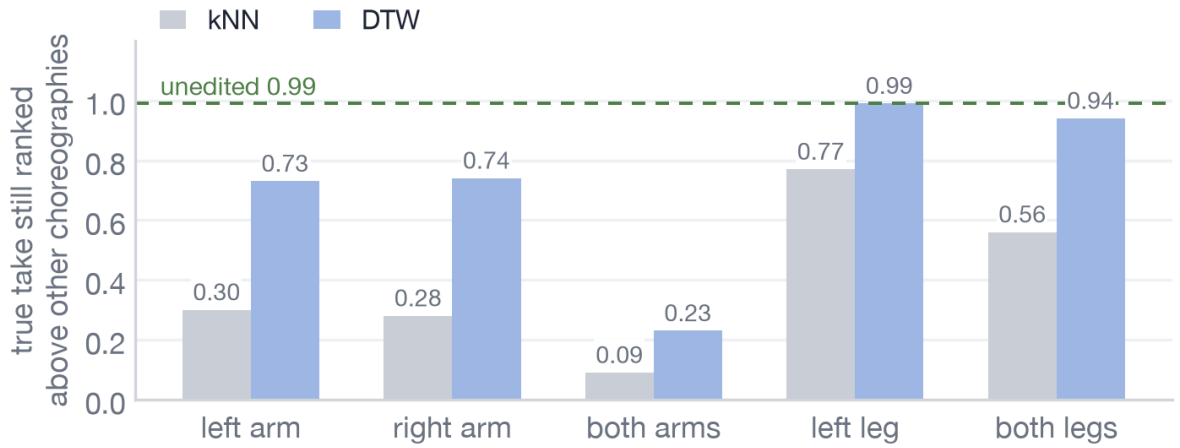


front and side camera

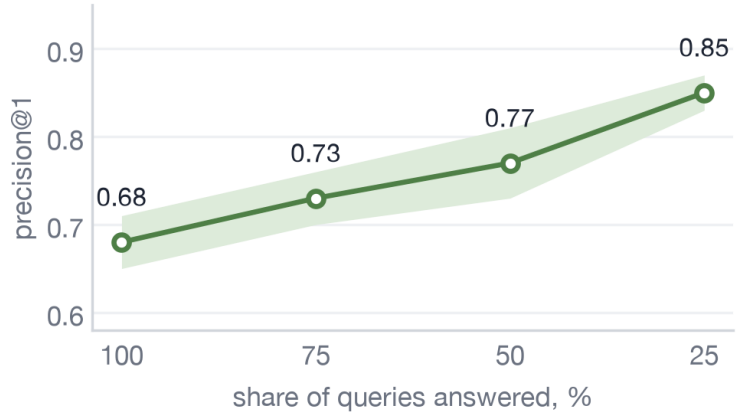
RESULT 2 · where the detail lives

Replacing one arm removes the match for kNN; abstaining on low confidence raises precision

REPLACE ONE LIMB BY ANOTHER CHOREOGRAPHY'S LIMB



ANSWER ONLY THE MOST CONFIDENT QUERIES

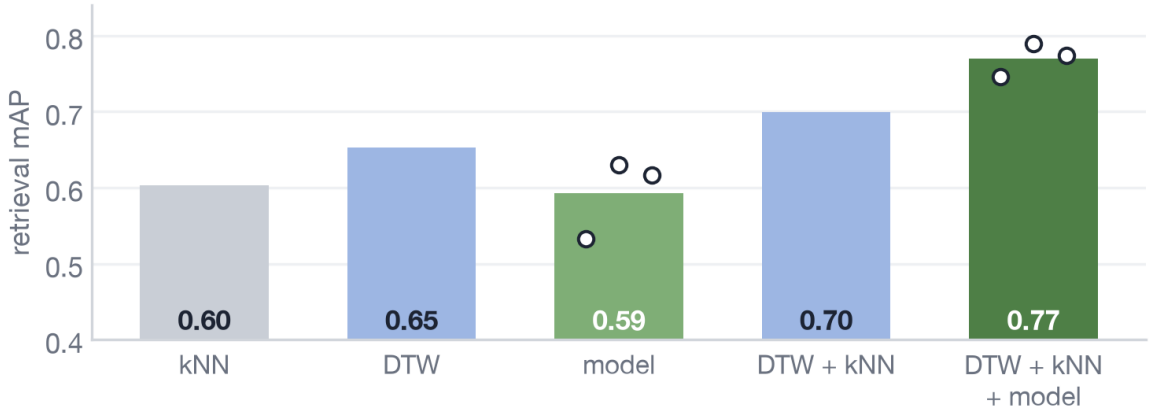


3 training seeds

RESULT 3 · the trained model, three seeds

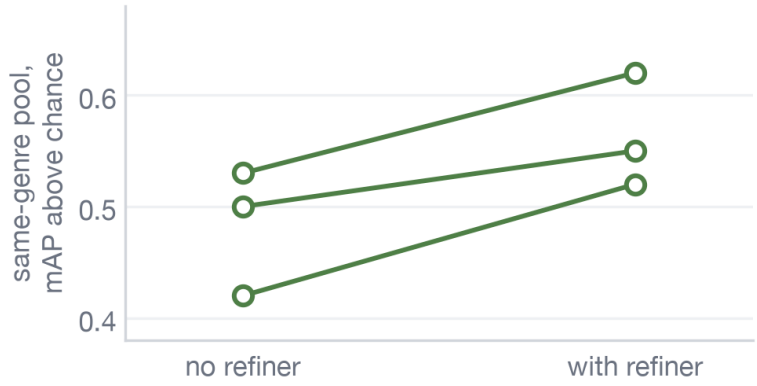
The trained model matches kNN alone; added to DTW + kNN it raises mAP from 0.70 to 0.77

MAP ON THE AIST++ CONFIRMATION SPLIT



circles = single training runs

REFINER ON SAME-GENRE NEGATIVES



one line per training seed

RESULT 4 · AI-generated video, with and without a library

Without a library of real clips AI-generated video is weakly detectable (AUC 0.53–0.76); with one, copies reach AUC 0.96–0.997

AUC BY SIGNAL



chance = 0.5 · text-to-video: 24 clips · copies: Wan-Animate re-dressings

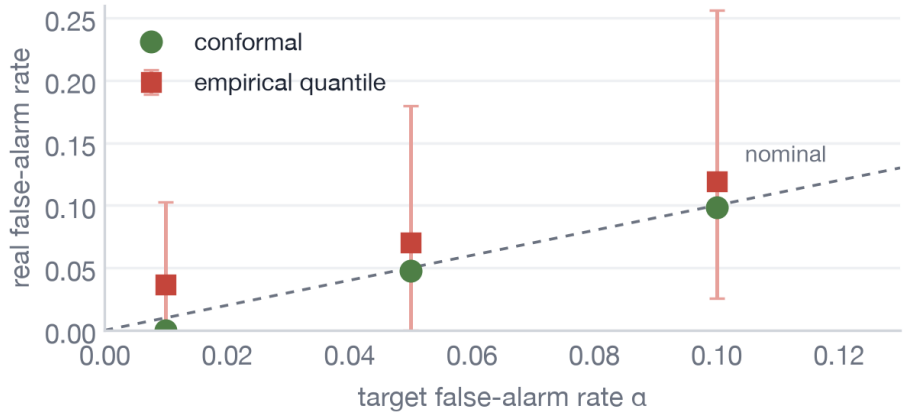
LEVEL 1 · no database: weakly detectable

LEVEL 2 · with a database: detects

RESULT 5 · calibrated false-alarm threshold

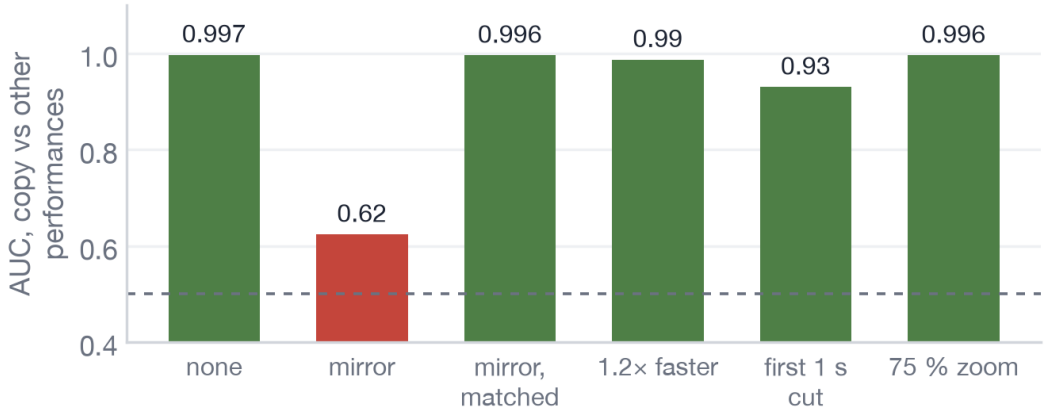
The empirical 1 % threshold gave 3.7 % false alarms; a conformal threshold holds the nominal rate

FALSE-ALARM RATE ON 2000 RANDOM SPLITS



78 real clips, half to calibrate, half to test

EDITED COPIES



AUC against other performances (AIST++, DTW)

GENERALIZATION · to any motion

The method assumes no choreography-specific label: only a skeleton and takes of the same motion

INPUT

Any skeleton split into parts (here 23 joints → 5 parts)

IDENTITY

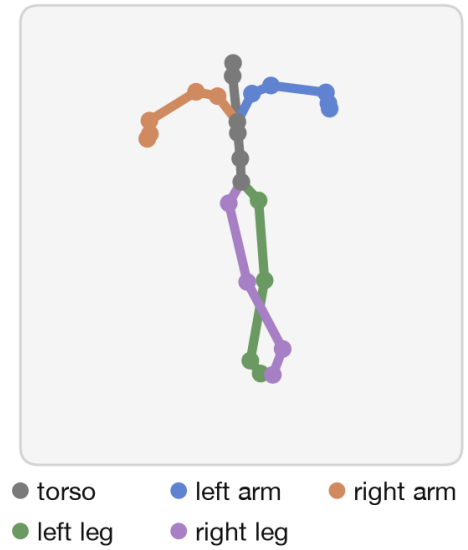
Positives P(i): the same motion from another take. No choreography-specific label.

COMPARISON

Part-wise cosines with visibility weights; camera-free through one similarity alignment A.

NEXT TEST

NTU RGB+D everyday actions



NEXT · what is still open

Open items: side view, offline pose estimator, direct-video baseline, calibration on the social media dataset

FOUND ON THE WAY

- **Visibility** does not drop when the arms are hidden
- **Path signatures** no gain (mAP 0.49)
- **One seed** is not enough (sd 0.04–0.08)

NEXT

1. **Side view:** retrain with the full yaw range
2. **Offline estimator** (WHAM, GVHMR) against CoMotion
3. **Direct-video baseline:** is a skeleton needed?
4. **Conformal threshold** on the social media dataset and edited copies